

FTD OS Company Compiler Full Flow and Engineering Report

Sole owner operating model and multi tenant provisioning architecture

Prepared for Alfreddie Postell II

Engineering context date 22 September 2026

Status Architecture direction and implementation plan

Executive decision

FTD_OS should use the Company Compiler as the single intake and planning system for every wholly owned company. The compiler may generate organizations with 1,000 or more logical positions, but it must separate organizational capacity from live provider accounts, paid licenses, service entitlements, and production authority.

Alfreddie Postell II is the sole ultimate owner and final human authority for the companies and assets in scope. Shared ownership, minority approval, joint venture consent, and outside equity approval are disabled by default. Company, tenant, credential, data, budget, and audit boundaries remain separate because ownership does not merge provider accounts or security boundaries.

The immediate engineering objective is a version controlled control plane that turns an approved COMPANY_BUILD_REQUEST into immutable plans, executes only tenant bound and human approved changes, verifies provider state, and preserves evidence and rollback data. The current workbooks and Apps Script files are planning sources and test fixtures. They are not production authorization.

Source plans and conclusions

Source	Scope	Value	Current classification
Hutchrok 1000 Agent Organization	1,000 positions, 15 departments, 45 teams, 85 groups, 6,045 memberships	Detailed organization and authority blueprint	Planning only
RunnerGang Provisioning Flow	Spreadsheet driven external membership import	Batching, retry, locking, resumability	Unsafe credential and tenant boundary
FTD Workspace Org Provisioner	Four parents, twenty companies, standardized nested groups	Declarative portfolio topology	No tenant proof, approval artifact, or rollback
Fee The Developer 1000 User Implementation	1,000 staged accounts, 2,000 aliases, five batches, 3,000 memberships	Useful generated execution package	License and conflict evidence still missing
Company Creation Intake	Twenty four intake sections and separate provisioning approval	Best candidate for canonical front door	Must compile into validated schemas

Chronological engineering direction

Order	Milestone	Engineering meaning
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Order	Milestone	Engineering meaning
1	Hutchrok organization model	Established a complete 1,000 position hierarchy with departments, teams, OUs, groups, reporting routes, approval tiers, and staged activation waves.
2	RunnerGang execution model	Added resumable batch processing but exposed privileged execution to editable spreadsheet state and lacked authoritative tenant binding.
3	FTD portfolio topology	Defined repeatable parent, company, and department group creation, but placed desired state directly inside executable source.
4	FTD 1,000 account package	Expanded the blueprint into user import staging, alias plans, group memberships, batches, and an administrative runbook.
5	Company Compiler intake	Generalized company creation into a reusable request contract that distinguishes unknown facts, generated structure, simulation, and external provisioning.
6	Sole owner decision	Set one ultimate owner and final human authority while preserving independent company and provider boundaries.
7	Current target	Build a reusable OS side compiler and control plane with plan, approve, apply, verify, audit, and rollback stages.

Sole owner accessibility model

The portfolio registry should declare Alfreddie Postell II as the 100 percent ultimate owner of every listed company and asset. This removes unnecessary co owner gates, but it does not justify shared credentials, merged tenant storage, or silent cross company access.

Control	Default	Effect
Ultimate owner	Alfreddie Postell II	Global portfolio visibility and final human approval
Ownership percentage	100 percent	No minority or partner approval workflow
Shared assets	Disabled by default	Asset use across companies requires an explicit service relationship
Agent self approval	Prohibited	No agent can approve its own restricted action
Privileged administration	Human only	Delegation remains capability specific and revocable
Owner override	Available and audited	Emergency action creates an immutable event and never removes evidence

Ownership and filing treatment

Disabled ownership flags. Shared ownership, minority consent, joint venture consent, outside equity approval, partner asset approval, and ownership dispute flags default to false.

Retained compliance flags. Formation records, annual reports, taxes, beneficial ownership reports where applicable, licenses, permits, insurance, payroll, regulated activity, government contracting, DBAs, and intercompany accounting remain subject to verification.

Evidence rule. The owner's statement is authoritative for internal design. External filings, domains, provider tenants, licenses, and permissions remain NEEDS_VERIFICATION until supported by authoritative records.

Accessibility hierarchy

Owner Control Plane

Company registry and global read visibility

Tenant access bindings

Company administrators

Human operators

Agent identities

Service identities

Approval and emergency suspension

Audit and resource metering

Core model distinctions

Object	Definition
Enterprise tenant	Legal or operating entity that owns resources
Workspace connection	Verified binding to one provider tenant using immutable provider identifiers
Company build request	Submitted intake, uploaded evidence, requested outputs, and unresolved information
Organization blueprint	Departments, teams, jobs, reporting routes, and logical positions
Logical position	An organizational slot that consumes no license by itself
Identity	A human, agent, service, contractor, partner, or simulation identity assigned to a position
Provider account	A real account created in Google Workspace or another SaaS provider
Capability	A narrowly defined operation the control plane may authorize
Entitlement	A provider authorized license, feature, quota, credit, or consumption right
Group	A communication or routing container that does not independently grant operational authority
Approval	Human authorization tied to an immutable plan and expiration
Audit event	Actor, tenant, capability, approval, request, result, and affected resource

Capacity model

The schema must support five, 1,000, or substantially more positions without changing its architecture. The value 1,000 is configuration. The compiler must calculate the following quantities independently.

- Logical capacity and reserved positions
- Active positions and assigned identities
- Human, agent, service, contractor, partner, and simulation identities
- Provider accounts required
- Paid licenses required and available
- Product entitlements and quotas
- Storage, API, model, execution, and spending limits

Complete Company Compiler flow

Step	Stage	Required action	Output
1	Receive intake	Human form, uploaded company package, internal blueprint, or approved prompt	COMPANY_BUILD_REQUEST draft
2	Normalize input	Validate field types, identifiers, ownership defaults, requested stage, and uploads	Canonical request
3	Classify evidence	Separate verified facts, owner assertions, generated structure, and missing external facts	Evidence register and NEEDS_INFORMATION list
4	Research gaps	Research only authorized missing facts and retain citations and dates	Research record
5	Resolve company identity	Assign company ID, legal and brand names, portfolio parent, ownership, and lifecycle state	Company registry entry
6	Select tenant strategy	Choose an existing tenant or propose a new isolated environment	Tenant assignment proposal
7	Verify domain and provider	Confirm customer ID, primary and secondary domains, license inventory, and administrative subject	Workspace discovery snapshot
8	Generate organization	Create departments, teams, jobs, reporting routes, and logical positions	Organization plan
9	Generate identity demand	Assign identity types and determine which positions require real provider accounts	Identity plan
10	Assign capabilities	Map jobs to capabilities, constraints, approval tiers, and separation of duties	Authorization plan
11	Inventory entitlements	Match identities and workloads to provider licenses, quotas, and service eligibility	Entitlement plan
12	Generate resources	Plan OUs, groups, aliases, mailboxes, service accounts, repositories, databases, and policies	Resource plan
13	Simulate	Test workflows, failures, approvals, capacity, and recovery without public activity	Simulation report
14	Produce provisioning diff	Compare desired state with live read only discovery using immutable IDs	Provisioning plan
15	Approve	Bind owner approval to tenant, plan hash, capability list, limits, and expiration	Approval record
16	Apply	Execute bounded, idempotent changes using a tenant specific connector	Execution record
17	Verify	Read provider state and reconcile every resource and membership	Verification report
18	Activate	Release only verified identities and capabilities for approved workloads	Activation record
19	Operate	Route work, meter consumption, monitor drift, and preserve audit events	Operational ledger
20	Recover or retire	Suspend, roll back eligible changes, revoke access, archive evidence, and release licenses	Rollback or retirement record

Canonical machine artifacts

Artifact	Purpose
company-build-request.json	Intake selections, owner assertions, uploads, requested outputs, and missing information
evidence-register.json	Source, authority, verification state, date, and applicable company
company-blueprint.json	Company identity, business model, tenant strategy, technology, and portfolio relationships
organization-plan.json	Departments, teams, jobs, reporting routes, and logical positions
identity-plan.json	Identity types, account requirements, aliases, managers, and activation state
authorization-plan.json	Capabilities, approval tiers, constraints, and separation of duties
entitlement-plan.json	Licenses, quotas, eligibility rules, allocations, and meters
workspace-discovery.json	Read only provider state with immutable customer and resource IDs
provisioning-plan.json	Exact proposed changes, dependencies, preconditions, operation ceiling, and rollback eligibility
approval-record.json	Owner, plan hash, tenant, approved capabilities, limits, expiration, and signature
execution-record.json	Idempotency keys, API results, checkpoints, failures, and created or adopted resource IDs
verification-report.json	Expected and observed state with unresolved drift
rollback-manifest.json	Compensating actions and before state for eligible changes

Every artifact requires a schema version, company ID, tenant ID, creation time, source commit, content hash, lifecycle state, and predecessor references. Spreadsheets and DOCX reports are generated views of these artifacts rather than execution authority.

Authorization and capability design

The control plane should authorize named capabilities instead of granting an identity general control. Owner access remains broad, but every action is still tenant bound, logged, and subject to provider permissions.

Capability	Meaning	Minimum gate
tenant.inspect	Read customer, domain, license, OU, group, and policy state	Read only
organization.compile	Generate organization and position plans	Compiler
deployment.upload	Upload a reviewed script or configuration version	Owner or deployment operator
ou.create	Create approved organizational units	Approved plan
group.create	Create approved groups	Approved plan
group.settings.change	Change visibility, posting, or membership policy	Separate approval
identity.create	Create a provider user or service identity	License proof and approval
membership.internal.add	Add an internal member	Approved plan

Capability	Meaning	Minimum gate
membership.external.add	Add an external member	Consent and separate approval
connector.install	Establish a provider connection	Tenant administrator approval
deployment.activate	Enable a production execution path	Owner approval
execution.rollback	Apply approved compensating actions	Owner or incident approval

Cross company access

Common ownership permits centralized visibility but does not create implicit access between companies. Any company providing engineering, operations, marketing, finance, or another service to another company should receive a governed service relationship.

- Provider company and consumer company
- Authorized service and exact resources
- Permitted capabilities and prohibited actions
- Data classification and storage boundary
- Budget or spending ceiling
- Effective date, expiration, and revocation state
- Owner approval and audit reference

OS build upload and execution architecture

The Windows workstation should host the developer interface, plan generator, validation tools, and reviewed deployment packages. It should not become the permanent vault for unrestricted multi tenant administrator credentials.

Windows CLI and repository

Intake validator and schema registry

Compiler and generated artifacts

Read only discovery adapters

Plan and approval service

Tenant specific execution queue

Apps Script adapter for limited bootstrap work

Direct Admin API adapter for durable operations

Verification, audit, metering, and rollback

Local command interface

Command	Purpose
inspect	Read current tenant and provider state without mutation
compile	Generate canonical plans from an approved build request
plan	Produce an exact desired versus observed change set
approve	Create a signed and expiring approval for one plan hash
upload	Upload reviewed code or configuration without activating it

Command	Purpose
apply	Execute only the approved plan within operation limits
verify	Re read provider state and reconcile the execution
status	Report execution, drift, license, and entitlement state
suspend	Stop a tenant queue or capability
rollback	Apply eligible compensating actions from the recorded manifest

Apps Script role

Apps Script may support tenant local bootstrap and limited administrative workflows. Source should be generated locally, reviewed in Git, uploaded as a version, and activated only through a separate approval. Editable spreadsheets must not control admin authorized background triggers. Long term multi tenant operations should use direct provider APIs through isolated connectors and a durable execution service.

Tenant expansion process

1. Register the company and proposed provider tenant without credentials.
2. Verify the displayed tenant identity, immutable customer ID, and primary domain.
3. Obtain read only discovery authorization from that tenant's administrator.
4. Import current domains, licenses, users, groups, OUs, policies, and connector state.
5. Compare discovery with the company blueprint and classify all differences.
6. Review required scopes, administrative subject, and capability limits.
7. Create an isolated tenant connection and secret binding.
8. Generate a zero write plan and verify the operation ceiling.
9. Obtain owner approval for a bounded pilot.
10. Apply the pilot, verify it, and confirm rollback evidence.
11. Expand in approved waves with independent checkpoints.
12. Retain a tenant specific revocation and emergency suspension path.

Status model

State	Meaning
DRAFT	Input may change and has not passed validation
NEEDS_INFORMATION	A required factual input or authoritative record is missing
VALIDATED	Schema and internal consistency checks passed
PROVISIONING_PLAN_READY	A reviewed change plan exists but has no execution authority
HUMAN_APPROVAL_REQUIRED	The next step affects an external system or protected capability
APPROVED_FOR_EXECUTION	A current approval is bound to the exact tenant and plan hash
EXECUTING	Approved changes are in progress
PARTIALLY_APPLIED	Some changes succeeded and reconciliation is required
VERIFIED	Observed provider state matches the approved plan
ROLLED_BACK	Eligible changes were reversed and verified

State	Meaning
BLOCKED	A required authority, provider state, or recovery condition prevents safe continuation

Current plan classifications

Asset	Classification	Reason
Hutchrok 1,000 agent workbook	PROVISIONING_PLAN_READY	License, recovery, pilot, and audit prerequisites remain open
FTD 1,000 user workbook	PROVISIONING_PLAN_READY	Dashboard wording is more optimistic than the unresolved gates
RunnerGang script	SECURITY REVIEW REQUIRED	Mutable sheet data can steer privileged trigger execution
FTD provisioner script	SECURITY REVIEW REQUIRED	No authoritative customer binding, approval record, or rollback
Company Compiler intake	VALIDATED DESIGN DIRECTION	Machine schema and implementation remain to be built

Material risks and required controls

Severity	Risk	Required control
CRITICAL	Wrong tenant execution	Bind every plan and credential to an immutable customer ID and fail closed
CRITICAL	Credential confused deputy	Never let editable workbooks steer background admin credentials
CRITICAL	Cross tenant privilege spread	Use isolated connectors, queues, secrets, logs, and revocation per tenant
HIGH	Unapproved mass account creation	Require license evidence, exact counts, plan hash, expiration, and owner approval
HIGH	Existing resource adoption	Verify immutable IDs, ownership, purpose, managers, and configuration before reuse
HIGH	Missing rollback	Record before state and compensating actions before each mutation
HIGH	Group based approval bypass	Evaluate capabilities and approval records independently of membership
MEDIUM	Quota and propagation failures	Use structured retries, checkpoints, readiness checks, and bounded waves
MEDIUM	Mutable or incomplete audit	Write append only external events before and after every operation
MEDIUM	Schema drift	Version schemas and migrations and preserve artifact lineage

Implementation roadmap

Phase	Objective	Scope	Exit condition
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Phase	Objective	Scope	Exit condition
Phase 0	Freeze unsafe production entry points	Retain read only discovery and dry run only	No uncontrolled writes
Phase 1	Schema foundation	Company, tenant, position, identity, capability, entitlement, approval, and audit schemas	Versioned contracts
Phase 2	Compiler	Convert intake into blueprint, organization, identity demand, and generated reports	Deterministic builds
Phase 3	Discovery	Read only Google Workspace and other provider adapters	Authoritative observed state
Phase 4	Planning and approval	Immutable diffs, policy checks, operation ceilings, and expiring approvals	Execution ready plans
Phase 5	Single tenant pilot	Group and OU creation for one verified tenant	Verified apply and rollback
Phase 6	Identity provisioning	Bounded user creation after license and recovery controls	Controlled activation waves
Phase 7	Multi tenant operations	Isolated connections, queues, metering, monitoring, and suspension	Portfolio control plane
Phase 8	Provider expansion	Additional SaaS entitlement and workload adapters	Governed cross provider routing

Tests required before production

Tenant and identity

- Wrong customer ID, domain, administrative subject, Cloud project, or deployment ID must fail closed.
- A manifest for one company cannot be paired with another tenant's credential.
- Revoked connections and changed administrator roles must stop new execution.

Plans and approvals

- Any change after planning invalidates the plan hash and approval.
- Expired, reused, broader, or unsigned approvals are rejected.
- Owner override is recorded and cannot erase prior evidence.

Execution

- Concurrent and repeated applies remain idempotent.
- Timeouts, quota limits, partial failures, and delayed provider propagation resume safely.
- Existing conflicting resources are not treated as harmless duplicates.

Recovery

- A partially completed wave can be reconciled and rolled back without removing preexisting resources.
- Audit storage failure stops mutation rather than producing an unaudited change.
- Emergency suspension prevents new operations without corrupting in progress evidence.

Scale and entitlements

- The compiler handles 5, 1,000, and larger configurations without schema changes.
- Logical positions do not consume provider licenses until an account is approved.

- Provider specific entitlement limits remain separate from identity counts.

Operational runbook

Order	Action	Requirement	Evidence
1	Inspect	Confirm company, tenant, customer ID, domains, licensing, connector identity, and current resources	Read only evidence
2	Compile	Generate canonical artifacts from the approved intake and evidence register	Versioned plan set
3	Review	Resolve NEEDS_INFORMATION and all policy violations	Validated inputs
4	Plan	Produce exact create, update, adopt, and no change operations	Immutable diff
5	Approve	Owner approves the tenant, plan hash, capability set, limits, and expiration	Approval record
6	Pilot	Apply the smallest useful wave	Pilot execution record
7	Verify	Reconcile provider state by immutable ID	Pilot verification
8	Expand	Apply later waves only after the previous wave is verified	Wave records
9	Activate	Enable workloads and entitlements approved for production	Activation record
10	Monitor	Track drift, consumption, failures, and expiring approvals	Operational ledger
11	Recover	Suspend, reconcile, or roll back using the execution manifest	Recovery evidence

Engineering decisions to preserve

- Freddie Postell II is the sole ultimate owner and final human authority for the portfolio in scope.
- All companies and assets remain separately addressable even when ownership is common.
- One owner receives unified visibility through explicit bindings rather than shared unrestricted credentials.
- Unknown external facts remain NEEDS_INFORMATION and are never invented.
- A logical position, identity, email address, alias, group, capability, and entitlement are separate objects.
- One thousand is a configurable capacity value rather than an architectural ceiling or automatic license count.
- Privileged administration remains human controlled.
- External execution and production activation require separate approval.
- Simulation activity remains separated from public customer activity.
- Every external mutation must be tenant bound, idempotent, verified, auditable, and recoverable.

Recommended next engineering package

The next implementation package should create the repository structure, JSON schemas, sample COMPANY_BUILD_REQUEST, portfolio registry, read only tenant discovery interface, plan format, approval record, and a command line prototype supporting inspect, compile, plan, and verify. Apply, upload, connector installation, and production activation should remain disabled until the tenant binding, approval, audit, idempotency, and rollback tests pass.

Source inventory

- Hutchrok_1000_Agent_Organization.xlsx dated 21 September 2026
- Fee_The_Developer_1000_User_Implementation.xlsx dated 22 September 2026

- RunnerGang_Provisioning_Flow.gs
- ftd-workspace-org-provisioner.gs
- Plays Ranch and FTD_OS Company Creation Intake Template
- Conversation decisions concerning authorized 1,000 seat capacity, sole ownership, accessibility, and multi tenant control